Day 127 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Find Quartiles and Interquartile Range

Standards: 6.SP.A.3, 6.SP.B.5c

Learning target: I can calculate quartiles and interquartile range.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Sort the data. Find the median, then the median of each half for Q1 and Q3. For an odd count, exclude the overall median from both halves in this course. IQR = Q3 - Q1; range = maximum - minimum.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references:

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01 / Learn

Sort the data. Find the median, then the median of each half for Q1 and Q3. For an odd count, exclude the overall median from both halves in this course. IQR = Q3 - Q1; range = maximum - minimum.

Worked example 1

Reading times in minutes are 2, 3, 3, 4, 4, 5. Find Q1, Q3, IQR, and range.

1. Sort the data.

2. Find the median, then the median of each half for Q1 and Q3.

3. Lower half [2, 3, 3] has median 3

4. upper half [4, 4, 5] has median 4.

5. Subtract Q1 from Q3.

6. State the result in the form requested: Q1 3, Q3 4; IQR 1 min; range 3 min.

Answer: Q1 3, Q3 4; IQR 1 min; range 3 min

Worked example 2

Six students recorded reading times (minutes): 5, 11, 11, 17, 17, 23. Part A: enter Q1 and Q3. Part B: enter IQR and explain which portion of the data it describes.

1. Sort the data.

2. Lower-half median is 11

3. upper-half median is 17.

4. IQR = 17 - 11 = 6, the span of the middle half of the ordered data.

Answer: Q1 = 11; Q3 = 17; IQR = 6 min

Day 127 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Find Q1, Q3, IQR, and range.

02 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Find Q1, Q3, IQR, and range.

03 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 8, 9. Find Q1, Q3, IQR, and range.

04 / Constructed response

Reading times in minutes are 5, 11, 11, 17, 17, 23. Find Q1, Q3, IQR, and range.

05 / Constructed response

Reading times in minutes are 2, 5, 5, 8, 8, 11. Find Q1, Q3, IQR, and range.

06 / Constructed response

Reading times in minutes are 2, 7, 7, 12, 12, 17. Find Q1, Q3, IQR, and range.

07 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Find Q1, Q3, IQR, and range. Explain what IQR describes.

08 / Multi-field numeric entry and explanation

Six students recorded reading times (minutes): 7, 11, 11, 15, 15, 19. Part A: enter Q1 and Q3. Part B: enter IQR and explain which portion of the data it describes.

09 / Multi-field numeric entry and explanation

Six students recorded reading times (minutes): 6, 11, 11, 16, 16, 21. Part A: enter Q1 and Q3. Part B: enter IQR and explain which portion of the data it describes.

10 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 6, 15. Find Q1, Q3, IQR, and range. Explain what IQR describes.

Day 127 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Find Quartiles and Interquartile Range

Standards: 6.SP.A.3, 6.SP.B.5c

Learning target: I can calculate quartiles and interquartile range.

1. Q1 9, Q3 10; IQR 1 min; range 3 min

Lower half [8, 9, 9] has median 9; upper half [10, 10, 11] has median 10. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

2. Q1 4, Q3 5; IQR 1 min; range 3 min

Lower half [3, 4, 4] has median 4; upper half [5, 5, 6] has median 5. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

3. Q1 7, Q3 8; IQR 1 min; range 3 min

Lower half [6, 7, 7] has median 7; upper half [8, 8, 9] has median 8. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

4. Q1 11, Q3 17; IQR 6 min; range 18 min

Lower half [5, 11, 11] has median 11; upper half [17, 17, 23] has median 17. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

5. Q1 5, Q3 8; IQR 3 min; range 9 min

Lower half [2, 5, 5] has median 5; upper half [8, 8, 11] has median 8. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

6. Q1 7, Q3 12; IQR 5 min; range 15 min

Lower half [2, 7, 7] has median 7; upper half [12, 12, 17] has median 12. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

7. Q1 6, Q3 8; IQR 2 min; range 12 min; IQR is the span of the middle half

Lower half [5, 6, 6] has median 6; upper half [7, 8, 17] has median 8. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

8. Q1 = 11; Q3 = 15; IQR = 4 min

Lower-half median is 11; upper-half median is 15. IQR = 15 - 11 = 4, the span of the middle half of the ordered data.

6.SP.A.3, 6.SP.B.5c | Multi-field numeric entry and explanation

9. Q1 = 11; Q3 = 16; IQR = 5 min

Lower-half median is 11; upper-half median is 16. IQR = 16 - 11 = 5, the span of the middle half of the ordered data.

6.SP.A.3, 6.SP.B.5c | Multi-field numeric entry and explanation

10. Q1 4, Q3 6; IQR 2 min; range 12 min; IQR is the span of the middle half

Lower half [3, 4, 4] has median 4; upper half [5, 6, 15] has median 6. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response